

Roy Einstein Mascarenhas

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SUMMARY

Principal Software Engineer with 12+ years of expertise in designing scalable backend systems using Java, Python, and cloud technologies like OCI, Azure, and GCP. Proficient in building microservices, event-driven architectures with Kafka, and leveraging SQL/NoSQL databases. Adept at performance optimization, CI/CD pipelines, and data science tools like Pandas and scikit-learn. Experienced in real-time systems, low-latency applications, and DevOps practices, delivering innovative solutions that drive business growth.

EDUCATION

Bachelor of Engineering in Computer Engineering - **Visvesvaraya Technological University**, Mangalore, India

June 2011

Courses: Algorithms, Computer Science, Data Structures, Database Systems, Cloud Computing, Distributed Systems, Operating Systems, Design Patterns, Microservices, Computer Architecture, Software Architecture, Computer Networks, Data Visualization Analytics, Big Data, Object Oriented Programming, Artificial Intelligence.

SKILLS

Languages: C/ C++, Python, JavaScript, Java, TypeScript, Go, Bash.

Frontend Frameworks: React.JS, OJET, React Native, Redux, Material-UI, HTML/HTML5, CSS/CSS3, Sass, Storybook, Bootstrap, Vue.JS, AJAX

Backend Frameworks: Node.js, Flask, Django, REST API, Jenkins, JSON, PANDAS, NumPy

Databases: SQL, SQLite, PL/SQL, PostgreSQL, NoSQL (Firebase, MongoDB), Elastic search, Kibana, Logstash

Cloud Platforms: Oracle Cloud Infrastructure (OCI), Microsoft Azure, GCP, AWS

Dev OPS: Docker, Jenkins, Shephard, Kubernetes, Terraform, Apache, Kafka, Celery

Developer Tools: Git, GitHub, GitLab, Postman, Linux/Unix, Figma, Adobe, VS Code, Apollo, Bitbucket, Confluence, Webpack, Jira (Scrum, Kanban)

Data Science/Data Engineering: scikit-learn, TensorFlow, PyTorch, PySpark, pandas, NumPy, AI/ML model training

EXPERIENCE

Principal Software Engineer - Oracle America, Las Vegas, NV

Mar 2023 - Present

- Spearheaded a team of 60 to develop and maintain Oracle Cloud infrastructure Compute Team.
- Conducted regular code reviews to ensure high-quality code and adherence to best practices.
- Mentored junior developers, providing guidance and support to help them grow their technical skills and advance their careers.
- Facilitated effective communication and collaboration within the team, fostering a collaborative environment, ensuring alignment with project goals and timelines.
- Played a key role in decision-making processes, influencing project direction and technology choices.

Senior Software Application Developer - Oracle India, Bengaluru, KA

May 2020 – Mar 2023

- Led a team of 25 to develop Scale Test Framework and RESTful API Frameworks. Designed high-availability API, UI, log retentions, Salt Stack integrated service to manage multi-VM connections, SUN test frameworks for test load check, CI/CD data pipelines, and customer service.
- Full-stack development in an application called Cloud Ruha. Designed framework for a self-service tool, developed using Python, Redis, Relational database, Pandas for data analytics, deployed under Kubernetes using Jenkins pipeline, container technologies.
- Conducted regular code reviews and provided mentoring to team members to improve coding standards and professional development.
- Enhanced team communication by organizing regular meetings, ensuring clear and open channels for feedback and updates.

Senior Software Full Stack Developer – Société Générale India, Bengaluru, KA

Oct 2018 – May 2020

- Led a team of 10 to develop RISK profile in a financial institution using advanced data gathering and analytics to produce readable One-click Action Applications, easing the management burden. Applications built using Python RESTful Framework (Flask), WebSocket, REDIS, MongoDB (secondary storage), Elastic Search, mathematical tools like Pandas, NumPy, and AI tools like Scikit for vector analysis.
- Deployed in a private cloud, UI designed and curated using Angular, based on test-driven development (TDD).
- Conducted code reviews to ensure high standards of software quality and to foster a culture of clean code.
- Mentored junior developers, helping them understand complex systems and improve their coding and problem-solving skills.
- Promoted effective communication within the team and with stakeholders to ensure project requirements and timelines were met.

Software Development Engineer – IBM INDIA Inc., Bengaluru, KA

Oct 2015 - October 2018

- Contributed to AIX operating (Linux Kernel/Unix) system patch development with the Hardware Management Console for physical devices. Improved 90% of operating system solutions in C++ and Python.
- Prepared development plans and patch codes to replace shell scripts with lightweight python-based modules. Troubleshooting, performance tuning, reducing code complexity, fixing code smell and blockers in SonarQube, and checking CI/CD pipeline development and code quality analysis.
- Facilitated clear and effective communication across the development team to ensure smooth and efficient project execution.

Software Development Engineer – Bally technologies (AKA Scientific games), Bengaluru, KA

June 2012 - October 2015

- Developed casino games for the casino industry using Objective C, C/C++, tested and deployed in Sandbox.
- Collaborated with the graphic team and project management team for requirement gathering, correction, and research development.
- Ensured effective communication with cross-functional teams to align project goals and deliverables.

PROJECTS

Fleet inconsistency demon - Java, Node.js, OCI, Jenkins, Docker, bitbucket, Shephard, Python, Kubernetes, Linux, Flask, Angular

- Developed a Demon app that manages stuck instances and workflow failures. Used Java, Node.js, integrated OCI worker nodes to scale resources based on traffic demand and set up a CI/CD pipeline using shepherd, Docker, and OCI to automate the build, test, and deployment for app delivery.
- Designed and developed a robust monitoring tool for Compute systems, ensuring high performance and scalability.
- Enhanced software architecture to meet evolving requirements, implementing scalable microservices using Kubernetes.
- Automated maintenance tasks with Linux shell scripts, streamlining processes and reducing system downtime by 25%.
- Led a team of developers, mentoring junior engineers to improve coding standards and performance optimization practices.

Automated Scale Test Framework – Salt-stack, Python, DOPE Ui, CSS, Node.js, OCI databases (No SQL), Redis, Pandas, Java, Kubernetes, React, Redis

- Designed a responsive backend app to test the instances before handover to the customer. Used DOPE to design UI/UX, react for frontend, and Node.js to handle the server side, integrated OCI notifications for real-time notifications, and utilized Python API to make HTTP requests to trigger the execution of OCI Workflows. Tool is capable of Stress / Performance testing on a scale of 1k instances in a given time.
- Built a scalable event-processing platform leveraging Kubernetes for container orchestration and load balancing.
- Developed server-side components in Python and Java, ensuring robust backend architecture to handle high traffic.
- Implemented client-side interfaces using React, improving user experience with real-time data visualization.
- Delivered project updates and insights in cross-functional meetings, aligning technical implementation with business goals.

Cloud Ruha – a self-service portal – Python, OJET, JavaScript, OCI, No SQL, MongoDB, Redis, Kubernetes, pandas, scikit-learn, TensorFlow, PyTorch, PySpark

- Engineered a Full stack app that handles all transactions such as Open search over the OCI resources to provide optimal selections, to cater all maintenance events on customer Instances such as PKI certs, upgrade DB, designed OJET UI, used Python, and Pandas, hosted on Oracle Cloud, integrated Oracle Multi-factor Auth for security, and utilized Nginx, Express for HTTP handling and server management, and MongoDB Atlas for database.

Resource Management – a data science tool – Python, MongoDB, Redis, Kubernetes, pandas, Open Search, scikit-learn , Flask, Angular, SQL, Linux

- Engineered a Data science tool that Curate the data sourced from KPI’s, stores in No SQL databases to cater Cloud Ruha application where decisions can be taken such as Orphan resources, unused resources, used Python, and Pandas curate the data, AI tools like scikit-learn to make the near to decisions, hosted on Oracle Cloud, integrated Oracle Multi-factor Auth for security, and utilized Nginx, Express for HTTP handling and server management.

Inventory and Capacity Management tool – a curated storage– Python, OCI, No SQL, MongoDB, Redis, Open Search

- Engineered a Management Backend tool that handles all transactions seamlessly as workflows, to cater all maintenance events and stores in databases such as PKI certs upgrade, used Python, and OCI workflow services (Wfaas), hosted on Oracle Cloud.

RISK Management tool – a financial decision maker– Python, Angular, No SQL, MongoDB, Redis, Private cloud, Elastic search, active directory, Open Search

- Engineered a RISK calculator tool, Used in Financial institute to identify potential threat in all Backend tools such as permission portal, customer data transactions, Orphan resource allocation, used Python, Pandas to curate the data and workflow services (Wfaas)to run the jobs, Elastic search to stores the key values hosted on Private Cloud, Redis used for caching technology, extensively used search algorithms
- Developed a full-stack application for financial risk analysis, combining Flask for backend services and Angularjs for responsive UI.
- Optimized SQL queries and database schemas to enhance data retrieval speed and accuracy.
- Designed shell scripts for automated deployment on Linux servers, ensuring efficient application management.
- Provided technical leadership, guiding the team in implementing scalable and maintainable solutions.

AIX Operating System – a Service packs– C++, Python, IBM sandboxes, GDB, Linux

- Part of the AIX OS service pack team to deliver new patches on field bugs reported by the Operating system using C++, Sandbox, IBM mainframes and Bare metals.

Casino games – C++, Sandbox, GDB, Shell scripting

- Part of Gaming studio and gaming app that runs in the Casino industry, use C++, and Sandbox.

Key Achievements

- Reduced search query latency by 35% through optimized storage and indexing.
- Improved search relevance and ranking accuracy, boosting user satisfaction by 30%.
- Achieved 25% cost savings by architecting efficient distributed storage systems.
- Processed 500M+ records daily in scalable search systems for enterprise applications.
- Mentored and led a team of 15 engineers, fostering a culture of innovation and technical excellence.

CERTIFICATIONS

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| <ul style="list-style-type: none">● Meta Front-End Developer● Microsoft Azure AI Fundamentals● Google Cloud’s Generative AI with Large Language Models | <ul style="list-style-type: none">● Oracle Cloud Infrastructure Foundations● Oracle Gen AI Certifications● DeepLearning.AI’s Neural Network & Deep Learning |
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